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Research Statement

Cosmologists have mapped the bookends of the Universe's history in remarkable detail. At one end, measurements of the Cosmic Microwave Background, have enabled precise constraints to be placed on its the initial conditions and physical processes governing its prior evolution. At the other, galaxy surveys have revealed the richly structured web of galaxies, filaments and voids that surround us today. Yet the timing and physics connecting these epochs remain poorly constrained, from the formation of the first stars and galaxies to their heating and ionisation of the surrounding gas. 21-cm cosmology offers a uniquely powerful route to reconstructing this history by measuring the weak radio signal from the surrounding neutral hydrogen, tracing its evolution and spatial fluctuations rather than individual sources.

However, realising this potential poses a notoriously challenging inference problem. The cosmological signal lies buried beneath contaminating emission four to five orders of magnitude brighter, and is observed through a complex instrumental response that strongly mixes contributions, rendering them highly degenerate in the observed data. Given the nuisance contributions cannot be sufficiently constrained a priori, 21-cm signal detection and characterisation is acutely sensitive to model misspecification. Reliable inference therefore demands a unified model of the full observational chain, spanning uncertainties in astrophysical emission, computational electromagnetics and electrical engineering.

End-to-end inference at the required scale and physical fidelity has previously been computationally intractable. My research tackles this problem by combining computational statistics, machine learning and distributed computing. I develop physically motivated, differentiable and hardware-accelerated generative models, alongside scalable Bayesian inference methods for exploring the high-dimensional, multimodal posteriors that arise. More broadly, I am motivated by the challenge of scaling statistical and numerical methods across physics, both algorithmically and computationally, to ever larger datasets, parameter spaces and physical systems.

As part of the Radio Experiment for the Analysis of Cosmic Hydrogen (REACH), I lead the development of its differentiable, hardware-accelerated Bayesian analysis pipeline, overcoming the computational bottleneck that limits the scale and fidelity of inference required for a trusted global cosmological detection. By reformulating the likelihood evaluation and integrating a nested-sampling implementation optimised for parallel GPU execution, my work has enabled numerous extensions to the modelling and Bayesian validation framework across the collaboration (Tutt et al., 2026a). One such extension is the integration of physically informed surrogate models for the complex electromagnetic response on the antenna. The central challenge is to construct a model expressive enough to capture physically plausible instrumental variation, yet sufficiently constrained by physics to preserve the identifiability of the cosmological signal. To address this, I developed a novel method for learning a latent representation of how the beam varies with geometry and environmental conditions from computational electromagentic simulations, compressing the beam parameterisation by two orders of magnitude and allowing the associated uncertainty to be analytically marginalised during inference (Tutt et al., 2026b).

Overall, the framework unifies astrophysics and computational electromagnetics through neural emulation, physics-informed parameterisations and latent-space representations. By reducing model evaluation from approximately 2.8 hours of wall time (1,608 CPU-core hours) to 0.14 milliseconds on a single GPU, a speed-up of 7.2×10^7, it provides a demanding scientific inference benchmark for sampling algorithms, simulation-based and variational inference (Tutt et al., 2026c).

For interferometric 21-cm cosmology, upcoming experiments such as the Square Kilometre Array bring a further challenge of scale, with exascale data-processing demands and models requiring millions of parameters. To meet this challenge, I have led the mathematical reformulation and computational scaling of BayesEoR, a Bayesian framework for joint signal and foreground inference. The central advance is to derive the prior covariance of the data by propagating physically motivated priors through a generative model of the measurement process. The resulting covariance captures high-dimensional correlations that are omitted by the phenomenological kernels of the Gaussian process approaches commonly used in this field. A noise-informed Karhunen–Loève expansion compresses this structure into a statistically optimal latent representation, controlling the discarded uncertainty relative to the observational noise. Constructed once and reused throughout inference, this representation amortises its construction cost and reduces the size of repeated matrix factorisations. For a realistic observational configuration, this collapses the joint parameter space from over 300,000 to under 10,000 and reduces the computational cost by three orders of magnitude, enabling posterior inference and Bayesian evidence evaluation in approximately 20 GPU-minutes (Tutt et al., in prep.).

Overcoming the bottleneck in BayesEoR led me to co-develop JAXMg with Roeland Wiersema from the Flatiron Institute's Center for Computational Quantum Physics. Working in different areas of physics, we faced the same need: solving large dense systems beyond the memory capacity of a single compute node. JAXMg adresses this by bring distributed linear solves and matrix decompositions directly into JAX, allowing these computations to scale across multiple GPUs and nodes within compiled scientific workflows (Tutt & Wiersema, 2026). With 22,000 downloads and 114 GitHub stars, JAXMg now supports research across computational physics, including the optimisation of neural-network representations of quantum many-body states within NetKet and tachys, two leading libraries in the field (Wiersema, 2026; Wan, Wiersema & Zhang, 2026). Ongoing development targets wider integration across scientific computing, including elliptic PDE solvers (Melia et al., 2025), computational electromagnetics (Gueuning et al., 2026) and Gaussian process inference (Pinder & Dodd, 2022).

My current work explores simulation-based inference across global and interferometric 21-cm cosmology, where increasing model dimensionality motivates moving beyond conventional posterior sampling. Drawing inspiration from field-level inference, I am investigating how to scale marginal neural posterior estimation to problems involving hundreds of thousands of nuisance parameters, amortising the need for repeated dense linear solves while allowing additional sources of uncertainty to be incorporated. Alongside this, I am exploring latent diffusion models and flow matching for the generation and super-resolution of three-dimensional cosmological fields, combining compressed representations with the flexibility to capture complex, non-Gaussian structure. These directions reflect the theme that runs through all of my research: scaling statistical and numerical methods without sacrificing rigorous uncertainty quantification. Whether through physically derived representations, distributed computation or learned approximations, my aim is to make principled inference tractable at scales previously out of reach, and to ensure that its conclusions can be trusted.

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